

Lecture: Convergence of Gradient Descent

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In the last lecture, we established that for convex functions, any critical point where $\nabla f(x) = 0$ is a global minimum. Today, we will show that gradient descent converges to the global minima for smooth convex functions.

1 Smoothness (Lipschitz Gradient)

While convexity prevents a function from having multiple disconnected minima, it doesn't prevent the function's curvature from being arbitrarily large. An algorithm can easily overshoot and diverge if the gradient changes too rapidly. *Smoothness* is a condition that limits how fast the gradient can change.

Definition 1 (Smoothness). *A function f is L -smooth for some $L > 0$ if its gradient is Lipschitz continuous with constant L . Formally, for all x, y :*

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2$$

This definition means the gradient cannot change arbitrarily fast. For twice-differentiable functions, smoothness is equivalent to having an upper bound on the Hessian, which means the function's curvature cannot be infinite.

Property 2 (Second-Order Condition for Smoothness). *A twice-differentiable function f is L -smooth if and only if its Hessian is bounded above:*

$$\nabla^2 f(x) \preceq LI \quad \text{for all } x$$

This means the largest eigenvalue of the Hessian is at most L .

An L -smooth function is quadratically upper-bounded. This is a crucial consequence that we will use in our proof.

Lemma 3 (Quadratic Upper Bound). *If f is L -smooth, then for all x, y :*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|_2^2$$

This inequality says that the function doesn't curve *upwards* too much faster than a quadratic with curvature L . It provides a global upper bound on the function's value.

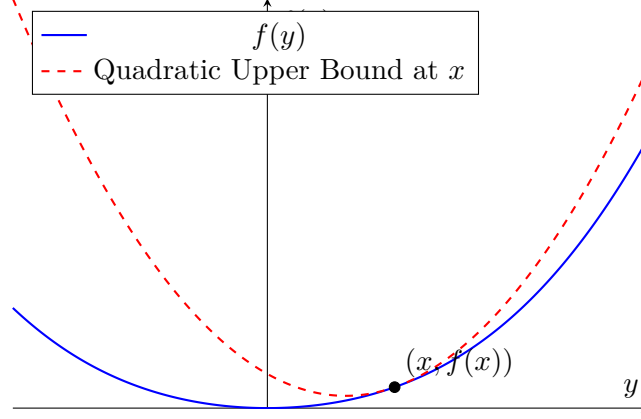


Figure 1: An L -smooth function is upper-bounded by a quadratic function at every point.

2 Convergence of Gradient Descent for Convex Functions

Recall the gradient descent algorithm:

Algorithm 1: Gradient Descent

Input: Starting point x_0 , step size η , number of iterations T

Output: Final parameters x_T

Initialize x_0 ;

for $t = 0, \dots, T - 1$ **do**

$x_{t+1} = x_t - \eta \nabla f(x_t)$;

end

return x_T ;

We have all the pieces to prove the convergence of gradient descent for functions that are both convex and smooth. We will prove the following theorem:

Theorem 4. *Let f be a convex and L -smooth function. If we run gradient descent with a fixed step size $\eta \leq 1/L$, then for any optimal point x^* :*

$$f(x_T) - f(x^*) \leq \frac{\|x_0 - x^*\|_2^2}{2\eta T}$$

This result shows that the “optimality gap” $f(x_T) - f(x^*)$ shrinks at a rate of $O(1/T)$. This means that to guarantee the error is less than some ϵ , we need to run for $T \geq \frac{\|x_0 - x^*\|_2^2}{2\eta\epsilon}$ iterations, so the number of iterations is $O(1/\epsilon)$.

2.1 Proof of Theorem 4

The proof rests on two key lemmas. The first, the **Descent Lemma**, guarantees that our function value decreases at each step. The second, the **Progress Lemma**, relates this decrease to the distance from the optimum. We will first prove these two lemmas and then combine them to prove the main theorem.

Lemma 5 (Descent Lemma). *For an L -smooth function, a gradient descent step with $\eta \leq 1/L$ guarantees that*

$$f(x_{t+1}) \leq f(x_t) - \frac{\eta}{2} \|\nabla f(x_t)\|_2^2.$$

Proof. We start with the quadratic upper bound from the smoothness property, letting $x = x_t$ and $y = x_{t+1}$:

$$\begin{aligned} f(x_{t+1}) &\leq f(x_t) + \nabla f(x_t)^\top (x_{t+1} - x_t) + \frac{L}{2} \|x_{t+1} - x_t\|_2^2 \quad (\text{smoothness upper bound}) \\ &= f(x_t) + \nabla f(x_t)^\top (-\eta \nabla f(x_t)) + \frac{L}{2} \|\eta \nabla f(x_t)\|_2^2 \quad (\text{substitute } x_{t+1} = x_t - \eta \nabla f(x_t)) \\ &= f(x_t) - \eta \|\nabla f(x_t)\|_2^2 + \frac{L\eta^2}{2} \|\nabla f(x_t)\|_2^2 \quad (\text{simplify inner product and norm}) \\ &= f(x_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(x_t)\|_2^2 \quad (\text{factor out } \eta \|\nabla f(x_t)\|_2^2) \end{aligned}$$

If we choose the step size $\eta \leq 1/L$, then $(1 - L\eta/2) \geq 1/2$, which gives the desired result. $\square$

Lemma 6 (Progress towards Optimum). *Let f be convex and L -smooth. For a gradient descent step with $\eta \leq 1/L$, we have:*

$$f(x_{t+1}) - f(x^*) \leq \frac{1}{2\eta} (\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2)$$

Proof. First, we analyze the change in distance to the optimum x^* :

$$\begin{aligned} \|x_{t+1} - x^*\|_2^2 &= \|x_t - \eta \nabla f(x_t) - x^*\|_2^2 \\ &= \|x_t - x^*\|_2^2 - 2\eta \nabla f(x_t)^\top (x_t - x^*) + \eta^2 \|\nabla f(x_t)\|_2^2 \end{aligned}$$

Using the first-order condition for convexity, $\nabla f(x_t)^\top (x_t - x^*) \geq f(x_t) - f(x^*)$, we can bound the middle term:

$$\|x_{t+1} - x^*\|_2^2 \leq \|x_t - x^*\|_2^2 - 2\eta(f(x_t) - f(x^*)) + \eta^2 \|\nabla f(x_t)\|_2^2$$

The Descent Lemma implies $\|\nabla f(x_t)\|_2^2 \leq \frac{2}{\eta}(f(x_t) - f(x_{t+1}))$. Substituting this in:

$$\|x_{t+1} - x^*\|_2^2 \leq \|x_t - x^*\|_2^2 - 2\eta(f(x_t) - f(x^*)) + 2\eta(f(x_t) - f(x_{t+1}))$$

The $2\eta f(x_t)$ terms cancel. Rearranging the remaining terms gives the desired result:

$$2\eta(f(x_{t+1}) - f(x^*)) \leq \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2$$

$\square$

With these lemmas in hand, the proof of the main theorem is straightforward.

Proof of Theorem 4. We sum the inequality from Lemma 6 over all iterations $t = 0, \dots, T-1$:

$$\sum_{t=0}^{T-1} (f(x_{t+1}) - f(x^*)) \leq \frac{1}{2\eta} \sum_{t=0}^{T-1} (\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2)$$

The right hand side is a telescoping sum that simplifies to $\frac{1}{2\eta}(\|x_0 - x^*\|_2^2 - \|x_T - x^*\|_2^2) \leq \frac{1}{2\eta}\|x_0 - x^*\|_2^2$. Since $f(x_t)$ is non-increasing (from the Descent Lemma), we have $f(x_T) \leq f(x_t)$ for all $t \leq T$. Therefore:

$$T(f(x_T) - f(x^*)) \leq \sum_{t=0}^{T-1} (f(x_{t+1}) - f(x^*)) \leq \frac{\|x_0 - x^*\|_2^2}{2\eta}$$

Finally, dividing by T gives the desired result.

$$f(x_T) - f(x^*) \leq \frac{\|x_0 - x^*\|_2^2}{2\eta T}$$

□

3 Faster Convergence under Strong Convexity

If we additionally assume the function is μ -strongly convex, we get a much faster rate.

Theorem 7. *Let f be μ -strongly convex and L -smooth. With $\eta = 1/L$, GD converges at a linear rate:*

$$\|x_T - x^*\|_2 \leq \left(1 - \frac{\mu}{L}\right)^T \|x_0 - x^*\|_2$$

This is called a *linear* convergence rate because the error is multiplied by a constant factor < 1 at each step, analogous to how a geometric series decreases. The error gets reduced by a constant factor every iteration, so the number of iterations to get ϵ error is $O(\log(1/\epsilon))$. This is much faster than the $O(1/\epsilon)$ rate for general convex functions.

While we will not see the general proof here, in the homework, we will show the result for the least squares problem.